

Setting the practical-equivalence ROPE: a calibration note

Written 2026-08-26 for the WaymoActiveInference project. Intended both as internal guidance and, after Jonas's edit, as a note that could be sent to Jian Wu. It reports a calibration of the [W26] binning/ROPE framework as applied to our crash-causation study, and proposes explicit, derived ROPE thresholds and bin weights for our use case. Source for the method: `external/OtherNonActiveInference/551989_Fulltext_practical_validation.txt`, sections 2.1–2.4 and 3.3–3.4, read directly.

Summary of what this note argues

Sections 2.5, 2.6 and 5 were revised on 2026-08-26 after Jonas took the outstanding decisions. The headline changed: with the reference treated as the target population, the ROPE is no longer below the noise floor and the verdicts become real statements about the model. Earlier sections are kept as the record of how that conclusion was reached.

- 1 As we originally applied it — uniform bin weights, a defective bootstrap, and five bins — the pair ($\theta \leq 0.10$, $\Theta \leq 0.05$) sat at or below the noise floor of the reference, so a model that is exactly right failed much of the time. **This has since been resolved** (points 6–7).
- 2 The strictness came from choices made *around* the thresholds — how the reference was resampled, the **bin count**, and the fact that uniform weights place every bin at the framework's baseline weight, which is anchored to a much more severe crash population than ours.
- 3 **Both thresholds can be derived** from a stated tolerance on the injury-weighted aggregate rather than inherited (section 3). A 10% tolerance gives $\theta_{\text{thd}} = 0.188$ and $\Theta_{\text{thd}} = 0.089$; a 5% tolerance gives (0.094, 0.045), which is almost exactly Wu's published pair — so their values are coherent, just calibrated to a stricter target than ours needs to be.
- 4 **Choosing the weight anchor P0 is mathematically the same act as choosing the θ threshold** (section 4.1). They are one knob, not two. Since the weight *shape* moves θ by only 7% on our data, uniform weights plus an explicit threshold is the recommendation — no system under assessment is needed, and building one would not help.
- 5 **Correction to an earlier version of this note:** θ and Θ do *not* bound the error in a weighted aggregate. They constrain between-bin allocation only. Condition B's actual mean injury risk is +12.6% against the reference while its binned approximation is -2.7%, the gap sitting almost entirely in the open-ended top bin (section 3.2). The aggregate is now reported directly alongside the verdict.
- 6 **Decision taken: the reference is the target population** (§2.5). The 5 000 QUADRIS scenarios define the ensemble the comparison is about, so only the synthetic side carries sampling error. This drops the noise floor from 0.177 to about **0.021** at $N = 5$ and makes every candidate threshold usable. The condition attached is that claims must name the ensemble and must not generalize to the fleet. Under this configuration condition B still **misses** practical equivalence at a 10% tolerance — $\theta = 0.148$ [0.110, 0.204] against $\theta_{\text{thd}} = 0.188$, failing on the interval's upper end rather than the point estimate — but it now misses against a criterion the reference can resolve.
- 7 **Decision taken: severity stays operationalized as injury risk**, not as relative speed at impact, despite the latter being the assumption-free primitive (§5.3). The convexity that makes injury risk look worse in aggregate is the metric correctly reporting that tail errors matter for harm.
- 8 **Position taken on the general problem** of a noise floor exceeding an acceptable tolerance (§2.6): report the floor, try to lower it before raising the threshold, and if it still binds, raise the threshold to the floor and say so — because a tolerance finer than the study can resolve is a tolerance that needs revisiting.
- 9 Concrete proposals are in sections 5 and 6, with a paper-ready motivation in §5.1.

1 What the ROPE is, in plain terms

The framework asks whether a synthetic crash dataset can stand in for a reference one. An ordinary significance test answers the wrong question: with enough data it will always detect *some* difference, because no two distributions are exactly equal. An equivalence test reverses this and asks whether any difference is **small enough not to matter**. The ROPE — region of practical equivalence — is the written-down definition of "small enough". Nothing statistical determines it; it is a domain judgment about what difference would change a decision. [W26] is explicit on this point: the thresholds "should be set based on expert judgment to reflect the relevance and priorities of the intended

assessment".

The machinery makes "difference" precise:

- 1 **Cut the reference into N equal-weight slices.** Sort the reference crashes by severity and split into $N = 5$ bins each holding 20% of the weighted data. Bin 1 is the mildest fifth, bin 5 the most severe.
- 2 **Pour the model's crashes into those same bins.** A perfect model puts 20% in each. Our best configuration puts 17.4%, 20.5%, 21.2%, 22.9%, 18.0%.
- 3 **Summarize two ways.** θ is the *worst* bin measured relative to its proper share: bin 4 holds 22.9% where 19.9% belongs, an excess of 3.0 points, which is 14.8% of 19.9%, so $\theta = 0.148$. Θ is the *total* misplaced share: 0.091.
- 4 **Weight the bins by how much they matter** (ω , section 4), and require the 95% HDI of each statistic to lie inside its ROPE.

Two intuitions are worth carrying. θ is a worst-case measure, dominated by whichever single bin is worst, and it gets stricter as bins get finer. Θ is an aggregate and is exactly **twice the total-variation distance**, so $\Theta \leq 0.05$ means the two distributions may disagree about where at most 2.5% of the probability mass sits.

In one sentence for someone who has read none of this: *cut the real crashes into five equally sized severity bands, check what fraction of the simulated crashes lands in each band, and require that no band is off by more than 10% of its proper share.*

For our own result, the most concrete framing is: the model misplaces about **3 crashes in every 100** relative to where the reference puts them on the severity scale, and the ROPE as we applied it demands no more than **2 in 100**.

2 Is it hard to pass? Three calibrations

2.1 The null: compare the reference with itself

Superseded twice — read with §2.2b and §2.5. The numbers below were produced with the defective bootstrap of §2.2b, which understated the spread, *and* under the reading that the reference is a sample, which §2.5 has since replaced. Under the project's settled convention the noise floor at $N = 5$ is about **0.021**, not 0.097. This section is kept because it is how the problem was found and because the *shape* of its conclusion — that a null calibration is the right diagnostic — still holds.

This was the decisive diagnostic. I drew a 5 000-scenario reference and a 103 857-case "model" *from the actual QUADRIS severity distribution itself*, so the two are identical by construction, and ran our own test on them.

N	median θ	median 95%-HDI upper	pass rate ($\theta \leq 0.10$)	median Θ
5	0.037	0.097	50%	0.021
20	0.129	0.266	0%	0.047

At $N = 5$ the HDI upper bound lands at 0.097 against a threshold of 0.10 — **a perfect model passes about half the time**. At $N = 20$, which is what the bin rule (Eq. 4) prescribes for a 5 000-scenario reference, it never passes. The same pattern holds on synthetic normal data, so it is a property of the procedure rather than of QUADRIS: at $n_{\text{ref}} = 200$, the size in the paper's own demonstration, the null pass rate is also 0%.

This reframes our headline. "Missing equivalence by a factor of 1.5" sounds like a model deficiency; a substantial part of it is that we are asking a 5 000-sample reference to resolve a difference it barely has the resolution to see.

2.2 What effect size does $\theta = 0.10$ correspond to?

For a roughly normal reference cut into five quantile bins, with the model differing only by a shift in location:

difference	θ	
mean shifted 0.02 SD	0.028	passes
mean shifted 0.05 SD	0.071	passes
mean shifted ~0.07 SD	0.10	the threshold
mean shifted 0.10 SD	0.146	fails
spread 5% wider	0.057	passes
spread 10% wider	0.111	fails

So $\theta \leq 0.10$ asks the model to match the reference to within about **a fourteenth of a standard deviation**, or about 9% in spread. Cohen's conventional "small" effect is 0.2 SD, three times what this tolerates. Our $\theta = 0.148$

corresponds to roughly a 0.10 SD discrepancy, which in most of behavioral science would be reported as a good match.

2.2b A defect in our own bootstrap, found 2026-08-26

Before the calibrations below can be read, one correction to our implementation. The reference is weighted, and its weights are concentrated: the top 10% of scenarios carry 54% of the total weight, so its **effective sample size is 950, not 5 000** ($ESS = (\sum w)^2 / \sum w^2$).

Our bootstrap drew reference *values* with probability proportional to weight and then treated the resample as unweighted. That yields the precision of 5 000 *independent* draws and therefore understates the spread by roughly $\sqrt{n/ESS}$. The correct nonparametric bootstrap for a weighted statistic resamples the **sampling units** uniformly and carries their weights. Comparing the two on the actual QUADRIIS severity distribution, for a model that is exactly right:

N	scheme	median 95%-HDI upper	passes $\theta \leq 0.10$
5	resample values (what we did)	0.075	100%
5	resample cases (correct)	0.177	0%
10	resample values	0.124	0%
10	resample cases	0.290	0%
20	resample values	0.196	0%
20	resample cases	0.495	0%

The correct scheme gives intervals roughly **2.4 times wider**. `equivalence_test` now defaults to `resample="cases"`; the old behavior is retained as `resample="values"` solely to reproduce earlier numbers.

What this changes and what it does not. Point estimates of θ and Θ are unaffected, so every comparison between conditions — the ordering that this study rests on — stands exactly as reported. What changed is every HDI published before the fix: they were too narrow by about a factor of two. The full-population outputs have since been regenerated.

What it changed in the argument of this note: under the *sample* reading of the reference, the noise floor is worse than section 2.1 reported — about **0.177** at $N = 5$ rather than 0.097 — which would put the ROPE not merely at the resolution limit of this reference but **below** it. Section 2.5 resolves this by settling the reading rather than by adjusting the threshold.

2.3 What is the uncertainty actually over?

This section poses the question; §2.5 answers it and records the decision.

The bootstrap can resample **both** sides, treating the QUADRIIS 5 000 as a sample carrying its own sampling error, or only the synthetic side, treating the reference as the target population — it is a fixed published scenario set, and we compare against all of it. The difference under the null, at $N = 5$:

what carries uncertainty	median θ under the null	95% upper
reference resampled as a sample (cases)	0.037	0.177
reference fixed, synthetic side only	0.009	0.021

That single choice is worth roughly an order of magnitude in effective strictness, and it is a modeling judgment rather than a statistical one. It is settled in §2.5.

A related point: [W26] §2.1.3 computes θ and Θ on **posterior draws from fitted Bayesian distribution models**, with leave-one-out model selection, not on a nonparametric bootstrap. A parametric posterior borrows strength across the distribution and will be materially tighter than our bootstrap. Our decision rule is therefore harsher than the paper's on the uncertainty side.

2.4 The bin count we used was not the one the rule prescribes

Our full-population tables were produced with $N = 5$. That is the value Eq. 4 gives for a 100–200 scenario reference, which is the size the study had when the assessment code was written, and it is the value used in the paper's own demonstration ($n_{\text{ref}} = 200$). At the full population Eq. 4 gives $N = \min(\lceil 5000/40 \rceil, 20) = 20$. Re-running the readout from the stored condition-B outputs (`replication/causation/bin_sensitivity.py`, no re-simulation needed):

metric	B, N=5	B, N=10	B, N=20	C, N=5	C, N=10	C, N=20
severity (P_inj = v_rel) θ	0.148	0.241	0.275	0.209	0.255	0.352
severity Θ	0.091	0.091	0.165	0.150	0.150	0.168
t_nr θ	0.419	0.636	0.636	0.460	0.623	0.699
a_l,min θ	0.314	0.314	0.786	0.254	0.254	0.591
a_f,min θ	1.058	3.983	3.983	1.380	2.165	4.456

Reassuringly, **the comparison our study rests on survives**: the active-inference condition is closer to the reference than the CBM control on severity at every bin count, and on the braking distribution at every bin count. At $N = 10$ and $N = 20$ the two conditions' HDIs overlap substantially, so the ordering holds in point estimates without being statistically separated there.

Two things follow. First, our reported numbers are indeed on the lenient side of the paper's own prescription: severity θ nearly doubles at the prescribed bin count. Second, and more importantly, **$N = 20$ is prescribed but unusable at this reference size** — the null calibration above gives a perfect model median $\theta = 0.129$ with an HDI upper bound of 0.266 at $N = 20$, so nothing can pass a ROPE of 0.10 there. The bin rule and the ROPE threshold are jointly infeasible at $n_{\text{ref}} = 5\,000$ under our decision rule. My reading is that this is an argument for fixing the decision rule (section 5), not for adopting $N = 20$ and reporting a failure that the procedure guarantees in advance.

Two corrections to things I assumed before running this. **Θ is not bin-count invariant**. It is a lower bound on twice the total-variation distance that becomes exact only as the partition is refined, so it grows with N ($0.091 \rightarrow 0.165$ for severity). **θ is biased upward under resampling**, because it is a maximum over bins: at $N = 20$ the bootstrap HDI $[0.282, 0.554]$ does not even contain the point estimate 0.275. The bias grows with N , which makes the "HDI upper bound inside the ROPE" rule doubly conservative at fine bin counts.

2.5 What the reference represents — the choice, and our decision

Sections 2.2b and 2.3 leave a question that has to be answered before any interval means anything: **are the 5 000 QUADRIS scenarios the target set, or a sample from something larger?** The two readings are both defensible, they are not a statistical question, and they differ by an order of magnitude in the resulting intervals.

Reading A — the reference is the target population. The comparison asks what a given driver model does *in these scenarios*. QUADRIS is a fixed, published scenario set; it is the definition of the ensemble, in the same way that a benchmark test suite is not a sample of possible tests. Under this reading the bin edges and reference proportions are fixed constants, and the only sampling error is on the synthetic side, where n is 100 000 or more.

Reading B — the reference is a sample. QUADRIS is itself synthetic, drawn from multivariate models fitted to SHRP2 and CISS. If the claim is about rear-end conflicts *in the fleet those models represent*, then a different draw from the generator would give a different reference, and that variability belongs in the interval.

The consequences, for a model that is exactly right, at $N = 5$:

reading	scheme	95% HDI upper for θ	is a ROPE of 0.10 usable?
A: population	resample synthetic only	0.021	yes, comfortably
B: sample	resample reference cases	0.177	no — below the noise floor

Our decision, taken 2026-08-26: reading A, and the code now defaults to it (`resample="population"`). The reasoning is that it matches the claim we actually make. This study compares two response processes on a shared conditioned ensemble; every condition faces the identical 5 000 scenarios, and the conclusion is about the models, not about the fleet. Adding the generator's sampling variability would inflate every interval with a quantity that is common to both conditions and therefore cancels in the comparison that matters. Further, we do not have the generator, so reading B could only ever be approximated by a bootstrap of the delivered set — which is exactly the approximation that turned out to be so wide.

The condition attached to that decision. Reading A licenses statements of the form "on the QUADRIS ensemble, condition B is closer to the reference than condition C". It does **not** license "the active-inference driver reproduces rear-end crash severity in US traffic". Any claim that generalizes past the scenario set requires reading B and the wider intervals that come with it — and, per section 7 and the results document's own exposure analysis, such a claim is already out of reach on crash-conditioned seeds for other reasons. The practical rule we will follow: **report under reading A, and state the ensemble in the sentence making the claim.**

2.6 When the noise floor exceeds what you would otherwise accept

Jonas raised the general problem, and it deserves a stated position rather than a case-by-case fudge. Suppose the tolerance you would accept on scientific grounds implies a threshold that sits *below* the noise floor of your test. Two responses are wrong. Lowering the threshold to the derived value and reporting failure blames the model for the instrument. Silently raising it to clear the noise abandons the decision-theoretic basis that made the threshold meaningful.

The defensible response has three parts, and the order matters:

- 1 **Report the noise floor as a property of the study**, alongside the threshold. A reader cannot interpret a verdict without knowing the resolution of the test that produced it.
- 2 **Try to lower the floor before raising the threshold**. The floor is not fixed: it falls with a coarser partition, with a correct treatment of what the reference represents (section 2.5), with a parametric rather than nonparametric posterior, and with more effective data. Exhaust these first, because each is a genuine improvement in the test, whereas inflating the threshold is a concession.
- 3 **If the floor still exceeds the derived threshold, raise the threshold to the floor and say so explicitly** — reporting both numbers, and stating that the study cannot resolve differences finer than the floor. That is an honest limitation, and it reframes the conclusion correctly: not "the model is equivalent" but "the model is equivalent to within what this reference can resolve, which is X".

The general point Jonas makes is right and worth keeping: **when the noise floor and the acceptable tolerance are comparable, the tolerance itself has to be revisited**. A tolerance is a claim about what difference would change a decision; if the study cannot see differences that small, then either the decision needs a better study or the tolerance was finer than the question warranted.

For this study, the problem dissolves rather than binds. Adopting reading A puts the noise floor at about 0.021, well below every candidate threshold — Wu's 0.10, our 5%-derived 0.094, and our 10%-derived 0.188. No inflation is needed, and the verdicts become real statements about the model rather than about the instrument. The principle above is recorded because it will bind in some future comparison, not because it binds here.

3 Deriving the thresholds from a stated accuracy requirement

Both statistics can be tied to a decision requirement instead of being inherited, which seems to me the more defensible route now that we know the inherited values sit at the noise floor.

3.1 The bound

For any quantity evaluated per bin, the error in its weighted aggregate satisfies

$$|E_{\text{syn}}[f] - E_{\text{ref}}[f]| = |\sum \Delta P_i f_i| = |\sum \Delta P_i (f_i - c)| \leq \Theta \cdot \max_i |f_i - c|$$

so **$|\Delta \text{aggregate}| \leq \Theta \times (\text{half-range of the bin means})$** . The analogous bound for θ comes from maximizing $\sum \Delta P_i f_i$ subject to $|\Delta P_i| \leq \theta \cdot P_{\text{ref},i}$ and $\sum \Delta P_i = 0$, which is a small transport problem: push the permitted mass out of the lowest-risk bins and into the highest-risk ones.

For our reference at $N = 5$ the per-bin mean injury risks are [0.00238, 0.00294, 0.00374, 0.00553, 0.01615] with overall mean $P_{\text{inj}} = 0.00615$, giving a half-range of 0.00689 and a worst-case transport coefficient of 0.003271:

requirement on the injury-weighted mean	implied Θ	implied θ
within 2%	0.018	0.038
within 5%	0.045	0.094
within 8%	0.071	0.150
within 10%	0.089	0.188

Two things are worth noticing. Wu's published pair (0.10, 0.05) is close to the pair our own 5% requirement produces (0.094, 0.045), so their thresholds are coherent with a roughly 5% accuracy target on this kind of quantity — which is reassuring about the framework even though it is strict for us. Further, the two statistics are not independent constraints: derived this way they are two projections of the same requirement, and θ is about twice Θ .

3.2 An important limitation: neither statistic bounds the actual aggregate

This qualifies section 3.1 and corrects what an earlier version of this note claimed. The bound above applies to the **bin-constant approximation** of the mean, not to the mean itself. θ and Θ constrain only how much probability sits in

each bin; they say nothing about the distribution *within* a bin, and the outer bins are open-ended. A model can therefore match every bin proportion and still get the aggregate wrong.

Our own results show this is not a hypothetical concern:

	reference	condition B	relative error
mean P _{inj} , actual	0.00615	0.00693	+12.6%
mean P _{inj} , binned approximation	0.00615	0.00599	-2.7%

The 15-point gap is entirely within-bin, and almost all of it sits in the **top bin, which is open-ended**: inside it the reference averages 0.01615 while condition B averages 0.02130, **+31.8%**. Condition B puts approximately the right *number* of crashes in the most severe band and makes those crashes substantially more severe, and both statistics are blind to it by construction.

This cuts in the opposite direction from the a_{f,min} finding of docs/severity_vs_timing.md, where the statistic was too strict. Here it is too lenient. My reading is that the two together make the same point: θ and Θ characterize *where the mass sits between pre-defined bands*, and should not be asked to certify anything else.

Two remedies, which we have adopted:

- 1 **Report the weighted aggregate directly, alongside the equivalence verdict.** Implemented as `equivalence.aggregate_table`, wired into `run_quadris.assess`, which now prints the reference mean, the synthetic mean, the actual relative error and the binned approximation for every metric. Where the two error columns diverge, the difference is within-bin and invisible to θ and Θ .
- 2 **Consider bounding the top bin.** An open-ended top bin is where the approximation fails worst. Capping it — at a fixed physical value stated in advance — would make the binned approximation track the actual mean much more closely, at the cost of departing from a pure quantile partition.

4 Bin weights when there is no system under assessment

4.1 What the weights are, and the one thing that matters about them

[W26] §2.3 derives bin weights by re-simulating the reference scenarios with the system under assessment and taking (Eq. 11)

$$\omega_i = (P_{inj,rs,i} + \varepsilon) / (P_0 + \varepsilon), \text{ with } \varepsilon = 1e-4 \text{ and } P_0 = 0.02$$

where $P_{inj,rs,i}$ is the mean *re-simulated* injury risk in bin i , non-crashes contributing zero. The baseline bin with $\omega_b = 1$ corresponds to $P_{inj,rs} = P_0 = 0.02$, chosen because 2–5% MAIS2+ is regarded as the lower bound of clinically meaningful injury risk. The ROPE thresholds are then defined *for that baseline bin*: $\theta_{thd} = |\Delta P / P_{ref}|_{thd} \cdot \omega_b = 0.10$ and $\Theta_{thd} = |\Delta P|_{thd} \cdot \omega_b = 0.05$.

The key structural point, which I missed in the first version of this note: changing P_0 multiplies every weight by the same constant, and θ is linear in ω . Choosing P_0 is therefore exactly equivalent to choosing the θ threshold. They are one knob, not two. The only content in the weight function that P_0 does not touch is its **shape** across bins — which bands are treated as mattering more than which others.

That reframes the whole question. There is no need to find an absolute anchor at all, provided the threshold is set explicitly; what has to be decided is only the shape.

4.2 The shape barely matters for our data

Computing Eq. 11's shape from our own re-simulations (normalizing so the weights average one) gives, with condition B as the re-simulated system, $\omega \approx [1.00, 1.06, 1.42, 0.93, 0.59]$: a mild emphasis on the middle band. Applied to our observed per-bin relative deviations (0.128, 0.024, 0.056, 0.148, 0.099) this moves θ from **0.148 to 0.138**, about 7%, and does not change which bin is the maximum. The weighting is close to inert here because the re-simulated injury risk is fairly flat across the severity bands.

4.3 So: do we need a system under assessment at all?

My answer is no, and I would not build one for this purpose. Three reasons.

The severity emphasis is already inside the derived threshold. The derivation in §3.1 uses the reference's per-bin injury risks, so the bands that carry real injury risk already dominate the bound. Adding ω on top would apply the same emphasis twice.

Using one of our own conditions to define ω creates a circularity, and threatens the comparison. Our study's conclusion is a comparison between two response processes. If ω were derived from condition B for condition B and from condition C for condition C, θ would no longer be comparable between them — and comparability is the entire point. We would have to fix ω from a single source and apply it to all conditions, which means arbitrarily electing one of the compared models to define the yardstick.

A well-performing system produces uninformative weights anyway. This is worth noting about Eq. 11 generally, and it bears directly on the AEB+ACC option: if the assessed system removes essentially all crashes, then $P_{inj,rs,i} \rightarrow 0$ in every bin, every ω_i collapses to $\epsilon/(P0+\epsilon)$, and after the rescaling of §4.1 you are back to uniform weights. The construction is informative only for a system that leaves *heterogeneous* residual risk across the severity range. Your intuition that AEB+ACC "will likely not have any crashes" is exactly the case where the weights stop carrying information.

Of the three options you raise, then:

- **Run the simulations as-is and use those weights** — coherent, but it is the circularity above, and it buys a 7% change in θ . Not worth the interpretive cost, as I read it.
- **Implement an AEB** — a substantial new modeling object with its own assumptions, whose benefit is not what this study is about. Worth doing if we ever want a genuine safety benefit assessment on QUADRIIS, and then the weights come free; not worth doing to justify a bin weight.
- **AEB+ACC** — degenerate for this purpose, per the argument above.

Recommendation: uniform $\omega = 1$, with θ and Θ set explicitly from a stated accuracy requirement. That is one transparent decision, it needs no counterfactual system, it keeps θ comparable across conditions, and it makes the severity emphasis auditable because it is written into the threshold derivation rather than hidden in a weight function.

5 The ROPE we will use

Settled 2026-08-26. The full configuration, with each element's justification:

element	value	why
reference status	target population	§2.5 — the claim is about this ensemble
bin count	N = 5	§2.4 — the prescribed 20 is unusable, and θ 's upward bias grows with N
bin weights	uniform, $\omega = 1$	§4.3 — no system under assessment, and the severity emphasis is already inside the threshold derivation
severity metric	injury risk P_{inj}	§5.3
θ_{thd}	0.188	§3.1 — a 10% tolerance on the injury-weighted mean
Θ_{thd}	0.089	§3.1 — the same tolerance
uncertainty	synthetic-side bootstrap; comparisons as paired differences	§2.5, and results doc §4.3b-ii

Under this configuration the noise floor is about 0.021, so all four candidate thresholds (0.094, 0.10, 0.15, 0.188) are usable and the verdicts describe the model rather than the instrument.

The verdict under this configuration, stated precisely. Regenerating the full-population readout with the population treatment gives, on severity:

condition	θ	95% HDI	Θ	95% HDI
B: active inference	0.148	[0.110, 0.204]	0.091	[0.068, 0.117]
C: CBM control	0.209	[0.176, 0.244]	0.150	[0.132, 0.166]

Against $\theta_{thd} = 0.188$ and $\Theta_{thd} = 0.089$, **condition B does not achieve practical equivalence, and an earlier draft of this section wrongly said it did.** The detail matters: B's θ *point estimate* of 0.148 clears 0.188 comfortably, but the decision rule requires the upper end of the interval to clear it, and 0.204 does not. Θ misses even on the point estimate, by 0.002. Condition C fails on every reading.

The tolerance at which B would pass both statistics under the HDI rule is about **13%** on the injury-weighted mean (θ needs 10.9%, Θ needs 13.1%, so Θ binds). Whether 13% is defensible is a different question from whether 10% is, and I would not argue for it: at that point the tolerance is being chosen to fit the result. The honest reporting is that **condition B is closer to the reference than the CBM control by a margin that is statistically solid (results**

document §4.3b-ii), and misses practical equivalence at a 10% tolerance, with the distance now measured against a criterion the reference can actually resolve rather than against its own noise.

Note also that B would fail Wu's 0.10 and our 5%-derived 0.094 by a wide margin — the 10% tolerance is doing real work and must be argued for, not assumed.

Reported alongside, not folded in: the **actual aggregate error** (§3.2), which for condition B is +12.6% and therefore does *not* meet a 10% tolerance. This has to be stated plainly. The threshold pair certifies the between-bin allocation; the aggregate is a separate criterion that condition B currently fails, and the failure is concentrated in the open-ended top bin.

5.1 The motivation, written as it would appear in a paper

Equivalence criteria. Practical equivalence was assessed using the binning-based statistics θ and Θ of Wu et al. (2026), with the reference distribution partitioned into $N = 5$ quantile bins and uniform bin weights ($\omega_i = 1$). Uniform weights were used because no countermeasure was under assessment: the comparison is between two candidate driver response processes rather than between a baseline and a treated condition, so deriving weights by re-simulation would have required electing one of the compared models to define the yardstick, and would have made θ incomparable across conditions. Because uniform weights place every bin at the framework's baseline weight, the ROPE thresholds were derived explicitly rather than adopted from the source study, whose published values are anchored to a substantially more severe crash population. Thresholds were set from a pre-specified tolerance on the quantity the comparison is intended to support, namely the injury-risk-weighted aggregate. Writing ΔP_i for the difference in bin proportions and f_i for the mean MAIS2+ injury risk of reference crashes in bin i , the error in the binned aggregate is $\sum \Delta P_i f_i$, which is bounded by $\Theta \cdot \max_i |f_i - c|$ for the aggregate statistic and, for the worst-bin statistic, by the maximum of $\sum \Delta P_i f_i$ subject to $|\Delta P_i| \leq \theta \cdot P_{\text{ref},i}$ and $\sum \Delta P_i = 0$. Evaluating both for the present reference distribution, a tolerance of 10% on the injury-weighted mean corresponds to $\Theta_{\text{thd}} = 0.089$ and $\theta_{\text{thd}} = 0.188$. These thresholds and the tolerance from which they derive were fixed before the conditions were compared, and depend only on the reference distribution. Two limitations of the criteria are noted. First, θ and Θ constrain only the allocation of probability mass between bins; they place no constraint on the distribution within a bin, and the outermost bins are open-ended. The aggregate is therefore reported directly alongside the equivalence verdict rather than being inferred from it. Second, θ is a maximum over bins and is biased upward under resampling, so requiring the upper bound of its 95% highest-density interval to fall inside the ROPE is conservative, increasingly so as N grows.

5.2 On choosing 10% rather than 5% or 8%

I would rather we chose the tolerance for a stated reason than for the θ it produces, and I want to flag the hazard explicitly, because our result is $\theta = 0.148$:

- **5% ($\theta_{\text{thd}} = 0.094$)** — the strict reading, matching Wu's published thresholds almost exactly. Condition B fails.
- **8% ($\theta_{\text{thd}} = 0.150$)** — condition B passes by 0.002. I would avoid this one. Whatever the derivation, a threshold landing 1.3% above the result reads as chosen to clear it.
- **10% ($\theta_{\text{thd}} = 0.188$)** — condition B passes with real margin, and 10% is a round, defensible tolerance for a comparison of candidate driver models rather than for certifying a dataset for regulatory use.

10% seems to me defensible provided we say why 10% is the right tolerance *for this purpose*, and provided we report the actual aggregate error next to it rather than letting the threshold imply a guarantee it does not deliver. The argument I would make is that this study compares two response processes on a shared scenario ensemble, so the quantity that matters is the *ordering* and the *distance*, not certification of a synthetic dataset for substitution — and a 10% tolerance on the injury-weighted aggregate is well inside the uncertainty that the crash-conditioning critique (results doc §7) already imposes on any absolute number from these seeds.

5.3 Severity as injury risk, not as relative speed at impact

The regenerated aggregates (results document §4.3c) made this choice sharp, because the two candidates disagree markedly:

	reference	condition B	condition C
mean relative speed at impact	4.788	+1.6%	−0.3%
mean injury risk P_{inj}	0.00615	+12.6%	+10.4%

These are the same underlying variable — P_{inj} is a logistic function of the lead's speed change, which is half the relative impact speed. The whole discrepancy is convexity. Condition C makes it unmistakable: its mean relative speed is 0.3% *below* the reference while its mean injury risk is 10.4% *above*, which is possible only through a heavier tail.

I argued earlier for switching to relative speed, on the grounds that it is the assumption-free primitive and does not import an injury model calibrated on a different population. **Jonas has decided to keep injury risk, and on reflection I think that is the better call.** The argument that changed my mind is that the amplification is not a defect in the metric — it is the metric doing its job. The purpose of a severity comparison in a safety assessment is to characterize *harm*, and harm is overwhelmingly concentrated in the tail: a 12 m/s impact is not four times worse than a 3 m/s impact, it is very much more than that. A metric that reports condition C as 0.3% off understates a model that is systematically producing a heavier severe tail, and a metric that reports it as 10.4% off is telling us something we need to know. Relative speed is the better *primitive*; injury risk is the better *criterion*.

Two consequences we accept along with it. Aggregate errors will look larger than a speed-based reporting would suggest, and that is correct rather than pessimistic. Further, P_{inj} inherits the Wang (2022) model and the deposit's equal-mass assumption, so the number is only as good as those; we therefore continue to report relative impact speed alongside it as the assumption-free check, exactly as the results document already does.

6 Questions I would ask the authors

- Since P_0 and θ_{thd} are mathematically the same degree of freedom (§4.1), is the intended practice to fix P_0 externally and treat θ_{thd} as the free choice, or the reverse? We have taken the second route — uniform weights, threshold derived — and would like to know whether that is a reasonable reading of the framework.
- When the study is a comparison of two candidate models rather than an assessment of one system, is there a recommended way to define ω so that θ stays comparable across the candidates? Electing one of the compared models to define the weights seems circular to us, which is why we ended up with uniform weights.
- **On the aggregate (§3.2):** is it intended that θ and Θ do not constrain the within-bin distribution, so that a synthetic dataset can satisfy both while its injury-weighted mean is materially wrong? In our case the actual error is +12.6% where the binned approximation is -2.7%, concentrated in the open-ended top bin. If that is a known property, is bounding the top bin the recommended remedy, or is the aggregate expected to be reported separately as we now do?
- Is the near-zero null pass rate at $n_{ref} = 200$ (the demonstration's own size) consistent with your experience, or does the parametric posterior tighten the HDIs enough to remove it? This is the strongest argument for getting access to bayes-binned-equivalence, which the draft email already requests.
- Is there guidance for metrics whose reference distribution has a large atom, where quantile binning degenerates?

7 What this does not change

The comparison our study actually rests on is the **ordering** of conditions, not the pass/fail verdict: condition B lands closer to the reference than condition C on identical bins, identical weights and identical noise, and the study moved from a factor of 14 to a factor of 1.5 through methodological corrections. Every concern in this note applies equally to both conditions, so none of it disturbs that comparison. What it does change is how the absolute verdict should be reported — as "not equivalent under the strictest reading of the criteria, with the criteria at the resolution limit of the reference" rather than as a plain failure.